

Aggressive Formation Tracking for Multiple Quadrotors Without Velocity Measurements Over Directed Topologies

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This article addresses the aggressive formation tracking problem of quadrotors without velocity measurements over directed interaction topologies. A distributed formation control scheme is proposed for a multiple-quadrotor system, which includes an outer-loop position controller to achieve the desired formation and an inner-loop attitude

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controller to regulate the attitudes. First, a second-order auxiliary system is designed to overcome the unmeasurable linear velocity. Next, the position controller is presented using only the position information under the directed topologies. Then, the geometric attitude controller is developed to carry out aggressive rotation maneuvers. Furthermore, an auxiliary system based on Lie algebra is proposed to eliminate the unreliable angular velocity feedback caused by disturbances and bias perturbations. Using Lyapunov stability theory on a cascaded system, it is rigorously proven that the proposed formation controller is asymptotically stable. Finally, the applicability and performance of the proposed control scheme are demonstrated by the numerical simulations.

I. INTRODUCTION

Recently, the formation tracking of multiple unmanned aerial vehicles (UAVs) has attracted a considerable amount of attention. It has been proven to be advantageous for the dangerous and tedious missions in military and civilian fields, such as aerial mapping, military surveillance, telecommunication relay, and precision agriculture [1], [2], [3], [4]. Quadrotors, with features like simplified aerodynamic mechanism and rapid maneuverability, serve as ideal aerial platforms that have been popularly exploited in the UAV formation [5]. However, they suffer from both their inherent underactuated dynamics and limited sensing capabilities. Therefore, the formation tracking of quadrotors with partly available states has been an extremely challenging problem.

The underactuated dynamics characteristics of quadrotors pose one major challenge for control system designs. In general, the quadrotor is propelled with the help of the four independent actuators. Then, its dynamics can be seen as the interconnection of underactuated position dynamics and fully actuated attitude dynamics [6], [7]. To this end, a typical hierarchical control strategy has been developed to decompose the quadrotor system into an inner/outer-loop architecture. The outer-loop controller is proposed to govern the formation and generate the geometric commanded attitude, and an inner-loop attitude controller is designed to track the target attitude. Since the configuration space of quadrotor attitude motion is not necessarily a Euclidean space, attitude control has been vastly studied using various attitude parameterizations, such as Euler angles [8], [9], quaternions [10], [11], and rotation matrices [12], [13], [14], [15], [16], [17]. There are substantial works based on the Euler angles since their representation is intuitive in a Euclidean space. However, they show singularities in expressing the attitudes of the quadrotor. To address this issue, quaternions are used to design attitude controllers, but they are ambiguous in representing the attitude of the quadrotor. In contrast, the geometric attitude controller has been developed directly using the rotation matrix on the special Euclidean group. Therefore, the geometric attitude representation can completely avoid the singularities and ambiguities related to Euler angles or quaternions. As a result, the geometric controller using rotation matrices can achieve a global and unique attitude representation. Furthermore, the geometric attitude control approaches over directed interaction topologies can be summarized into the

local or the almost-global controller [18]. Compared with the local controller, the almost-global controller can offer a convergence region that constitutes the whole state space except for a “small” set of measure zero.

An important assumption in the aforementioned studies is the availability of velocity information for feedback. Unfortunately, this requirement is not always satisfied for quadrotor formation due to the limited communication resources, inaccurate velocity measurements, and sensor failures. On the other hand, the low-cost quadrotors might not be equipped with velocity sensors to save space, cost, and weight. To this end, it is meaningful to design control algorithms without velocity measurements. For one thing, the formation tracking for a group of quadrotors without linear information has been extensively investigated. Fan et al. [19] constructed a dynamic observer to estimate the unmeasurable linear velocities. The formation control approach using the distributed filter was proposed in [20] to deal with the unavailability of linear velocity feedback and communication. In [21], [22], and [23], the formation control algorithms for quadrotors without linear velocity feedback were proposed based on auxiliary systems. The case with a directed graph was studied in [24]. The case with delayed communication was studied in [25]. For another, angular velocity for feedback is rarely considered. In the literature, different angular-velocity-free control techniques were applied for a single underactuated rigid body. In [26], a reduced-order observer was designed using quaternions to eliminate the requirement of angular velocity. As an extension, the dual quaternions were explored to develop the controller in [27]. Several researchers [28], [29] have proposed full-order observers to estimate the angular velocity information. The case with fixed-time control was presented in [30]. In [31], an auxiliary dynamical system was designed to generate the necessary damping without angular velocity feedback. Remarkably, Miao et al. [32] and Sun et al. [33] addressed the formation tracking problem for quadrotors in the absence of linear and angular velocities. In almost all the aforementioned studies, the formation controllers were proposed using undirected graphs, which means that the communication among quadrotors is bidirectional, and the graph Laplacian possesses the symmetric positive-definite property. As a result, the reliability of formation tracking is decreased for a group of quadrotors due to the limited sensing and communication resource. Compared to the undirected interaction topologies, the directed interaction topology consumes only half of the resources. Currently, the velocity-free formation control of linear systems [34] and fully driven systems [35] under a directed interaction topology were studied. But the design of velocity-free controllers is considerably difficult for underactuated systems due to the strongly coupled dynamics. Therefore, the formation control of the quadrotors without velocity measurements is still a valuable and open research issue.

In this article, we study the formation tracking of quadrotors without velocity measurements under directed graphs. To overcome the lack of linear/angular velocity feedback for quadrotor formation, two auxiliary dynamic

systems driven by proper inputs are associated with each quadrotor to recover and generate the necessary damping. Hence, the reliability against limited sensing resources and sensor failures, in practice, is well enhanced. Compared with the available quadrotor formation tracking control schemes, the main contributions of this article can be summarized as follows.

- 1) The proposed formation scheme focuses on the velocity-free control case under directed topologies and demonstrates system stability under asymmetric graph Laplacian. However, the linear and angular velocities were assumed to be known in [8], [9], [10], [11], [12], [13], [14], [15], [16], and [17], the angular velocity feedback was needed in [19], [20], [21], [22], [23], [24], and [25], and the positive-definite property of the undirected graph was adopted in [32] and [33]. Compared to undirected graphs, directed graphs are a more universal interaction topology. Thus, the proposed approach can effectively reduce the requirements for communication bandwidth and connectivity in the multiple UAV formations.
- 2) A velocity-free formation controller is designed for underactuated multiple-quadrotor systems under directed graphs. Compared to formation tracking for linear systems [34] or fully actuated systems [35], the designed controller for underactuated systems is more challenging due to the features of high nonlinearities and couplings in both the translational and rotational motions. The high-order derivatives of position and attitude cannot be obtained simply by the difference method due to the presence of noise. Hence, a second-order auxiliary dynamics system is used to overcome the lack of the linear and angular velocity feedback. Compared with the observer-based control scheme in [19], the proposed method depends on only its current position information and the position of its neighbor robots without the exact object model.
- 3) An aggressive geometric attitude controller is designed for each quadrotor to recover and generate the necessary damping. Different from the angular-velocity-free local attitude controllers using the real-valued error [31], [32], the proposed attitude controller can achieve an almost-global convergence region and carry out aggressive rotation maneuvers, such as large initial angles. Furthermore, an auxiliary system based on Lie-algebra-valued functions is designed to eliminate the unreliable angular velocity feedback. Owing to the inevitable topological obstructions, the proposed attitude controller can be considered as the strongest possible controller in terms of convergence range.

The rest of this article is organized as follows. In Section II, the quadrotor system modeling and the statement of formation tracking problem under directed graphs are illustrated. Section III gives the formation control design and the stability analysis of the overall closed-loop system.

Simulation results for illustrating the performance and effectiveness of the proposed formation scheme are shown in Section IV. Finally, Section V concludes this article.

II. PROBLEM FORMULATION

A. Notation and Math Preliminaries

In this section, several notations and preliminaries on Lie algebra and graph theory are first recalled. Let 0_n and I_n denote the zero and identity matrices of the dimension n , respectively. For a matrix $B \in \mathbb{R}^{3 \times 3}$, the minimum and maximum eigenvalues are represented as $\lambda_{\min}(B)$ and $\lambda_{\max}(B)$, respectively, and the determinant of B is denoted as $\det(B)$. Denote $\otimes$ as the Kronecker product. The special Euclidean group is denoted as $SE(3)$. It is the semidirect product of the orthogonal group $SO(3)$ and $\mathbb{R}^3$. In addition, $SO(3)$ is the set of all orthogonal 3×3 matrices defined as

$$SO(3) = \{R \in \mathbb{R}^{3 \times 3} : R^T R = R R^T = I_3, \det(R) = 1\}$$

where $SO(3)$ is a Lie group. It is not only a differentiable manifold but also a matrix group. The Lie algebra $\mathfrak{so}(3)$ associated with $SO(3)$ is given by

$$\mathfrak{so}(3) = \{B \in \mathbb{R}^{3 \times 3} : B = -B^T\}$$

where $\mathfrak{so}(3)$ is the real vector space of 3×3 skew-symmetric matrices, and isomorphic to $\mathbb{R}^3$ in terms of vector space. The isomorphism operator $(\cdot)^\wedge : \mathbb{R}^3 \mapsto \mathfrak{so}(3)$ is defined such that $(x)^\wedge y = x \times y$ for all $x, y \in \mathbb{R}^3$, where $\times$ denotes the cross product. The inverse of $(\cdot)^\wedge : \mathfrak{so}(3) \mapsto \mathbb{R}^3$ is denoted by $(\cdot)^\vee$. For $\forall x \in \mathbb{R}^3$ and $B \in \mathfrak{so}(3)$, we have $((x)^\wedge)^\vee = x$ and $((B)^\wedge)^\vee = B$. The matrix exponential $\exp(\cdot)$ represents an analytic diffeomorphism between $U_{\mathfrak{so}(3)} := \{\omega^\wedge \in \mathfrak{so}(3) : \omega \in \mathbb{R}^3, \|\omega\| < \pi\}$ and $U_{SO(3)} := \{R \in SO(3) : \text{trace}(R) \neq -1\}$. The principal matrix logarithm is used to denote the inverse map from $U_{SO(3)} \rightarrow U_{\mathfrak{so}(3)}$. It is defined as follows:

$$\text{Log}(R) = \sum_{i=1}^{\infty} \frac{(-1)^{i+1}}{i} (R - I_3)^i. \quad (1)$$

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote a **directed graph** associated with a group of n UAVs, in which $\mathcal{V} = \{v_1, v_2, \dots, v_n\}$ denotes the finite nonempty node set and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ denotes the edge set. In the directed graph, an edge $(v_i, v_j) \in \mathcal{E}$ denotes that UAV j can receive information from UAV i through sensing or communication. UAV j is the child node, while UAV i is the parent node, but not vice versa. The directed path from drone i to drone j is a series of edges. Generally speaking, a directed tree means that in addition to the root node, each node has only one parent node and the root node has a directional path to every other node. The directed spanning tree of a directed graph is a direct tree that contains all the nodes of the directed graph. The adjacency matrix $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{n \times n}$ associated with the directed graph $\mathcal{G}$ is defined such that if $(v_j, v_i) \in \mathcal{E}$, we have $a_{ij} > 0$; otherwise, $a_{ij} = 0$. In this article, self-edges are not allowed, i.e., $a_{ii} = 0$. The Laplacian matrix $\mathcal{L} = [l_{ij}] \in \mathbb{R}^{n \times n}$ is $\mathcal{A}$ is defined as $l_{ij} = -a_{ij}$, $i \neq j$ and $l_{ii} = \sum_{j=1, j \neq i}^n a_{ij}$.

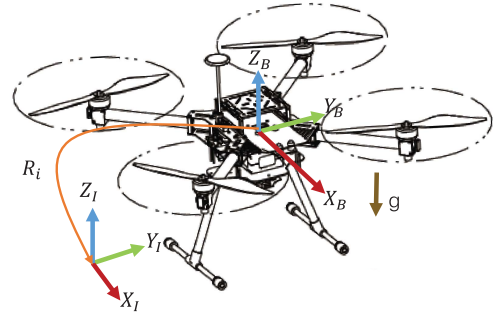


Fig. 1. Diagram of a quadrotor.

In the stability analysis of the multivehicle system, there are some useful lemmas that will be needed.

LEMMA 2.1 (See [36]) Let $\mathcal{G}$ be a direct graph of order n , and $\mathcal{L} = [l_{ij}] \in \mathbb{R}^{n \times n}$ is the associated Laplacian matrix. **The Laplacian matrix $\mathcal{L}$ has a single zero eigenvalue, and all other eigenvalues have positive real parts if and only if the interaction topology $\mathcal{G}$ has a directed spanning tree.**

LEMMA 2.2 (See [37]) Consider the cubic polynomial

$$F(\lambda) = \lambda^3 + a\lambda^2 + (a_1 + jb_1)\lambda + a_2 + jb_2 \quad (2)$$

where $j = \sqrt{-1}$, and a, a_1, a_2, b_1 , and b_2 are real numbers. Then, all the three roots of $F(\lambda) = 0$ have negative real parts if and only if $a > 0$, $a^2 a_1 - aa_2 - b_1^2 > 0$, and $a_2(a_2 - aa_1)^2 - b_1^2(a_1 a_2 + b_1 b_2) + ab_1 b_2(aa_1 - 3a_2) - a^3 b_2^2 > 0$.

LEMMA 2.3 (See [32]) Consider the second-order system as

$$\ddot{x} = -k_1 \sigma(\dot{x}) - k_2 \sigma(x) + o(t) \quad (3)$$

where $x = [x_1, x_2, x_3]^T \in \mathbb{R}^3$, and $k_1 > 0$ and $k_2 > 0$ are positive gains. If the continuous function $\|o(t)\|$ is bounded and tends to zero as $t \rightarrow \infty$, and the function $\sigma(x) = [\sigma(x_1), \sigma(x_2), \sigma(x_3)]^T$ satisfies the following:

- 1) $\sigma(0) = 0$, and $x_i \sigma(x_i) > 0$ for any $x_i \neq 0$, and $\exists \bar{\sigma} > 0$ such that $|\sigma(x_i)| < \bar{\sigma}$, where $i \in \{1, 2, 3\}$;
- 2) $\exists \sigma_r > 0$, the derivative of $\sigma(x_i)$ satisfies $|\dot{\sigma}(x_i)| \leq \sigma_r$;

then $\|\dot{x}\|$ and $\|x\|$ are bounded and tend to zero as $t \rightarrow \infty$.

LEMMA 2.4 (See [22]) Let $V : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ be continuously differentiable, and $W : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ uniformly continuous satisfying, for each $t \geq 0$, that

$$\dot{V}(t) \leq -W(t) + \gamma(t)\sqrt{V(t)} \quad (4)$$

where the nonnegative parameter $\gamma(t) \rightarrow 0$; then, $V(t)$ is bounded, and $W(t) \rightarrow 0$ as $t \rightarrow \infty$.

B. System Modeling

Consider a group of multiple-quadrotor systems with the leader-following architecture; suppose that there exist n followers, labeled as UAV 1 to n , and a virtual group leader

labeled as UAV 0, in a team. As shown in Fig. 1, each follower quadrotor is an underactuated rigid body. It can rotate around the center of gravity in the 3-D space without translation. Exploiting the Euler–Lagrange formulation, the dynamics of each quadrotor is presented in the following form:

$$\begin{aligned}\dot{p}_i &= v_i \\ m_i \dot{v}_i &= f_i R_i e_3 - m_i g e_3 \\ \dot{R}_i &= R_i \Omega_i^\wedge \\ J_i \dot{\Omega}_i &= \tau_i - \Omega_i^\wedge (J_i \Omega_i)\end{aligned}\quad (5)$$

where $i \in \mathcal{N} = \{1, 2, \dots, n\}$, m_i denotes the i th quadrotor mass, $J_i \in \mathbb{R}^{3 \times 3}$ denotes the symmetric and positive-definite inertia matrix, and $p_i \in \mathbb{R}^3$ and $v_i \in \mathbb{R}^3$ denote the position and linear velocity with respect to the earth-fixed inertial frame, respectively. Let $\Omega_i \in \mathbb{R}^3$ denote the body angular velocity, and the rotation matrix $R_i \in \text{SO}(3)$ can transform the body reference frame $\mathcal{F}_B$ to an inertial frame $\mathcal{F}_I$. The inputs f_i and $\tau_i \in \mathbb{R}^3$ denote external thrust and torque expressed in its body frame.

Here, a directed graph $\bar{\mathcal{G}} = \{\bar{\mathcal{V}}, \bar{\mathcal{E}}\}$ can be used to describe the directed interaction topologies between a virtual leader and followers with the node set $\bar{\mathcal{V}} \triangleq \{0, \mathcal{V}\}$ and the edge set $\bar{\mathcal{E}} \subseteq \bar{\mathcal{V}} \times \bar{\mathcal{V}}$. Here, $a_{i0} > 0$ if the virtual leader is a neighbor of follower i , and $a_{i0} = 0$ otherwise. Denote matrix $\mathcal{D} = \text{diag}(a_{10}, a_{20}, \dots, a_{n0})$ for ease of notation. Therefore, the network topology of leader–follower formation under directed graph $\bar{\mathcal{G}}$ can be presented by the interaction matrix $\mathcal{H} = \mathcal{L} + \mathcal{D}$.

C. Problem Statement

The main goal of this article is to drive a multi-quadrotor system (1) to achieve aggressive formation tracking without both the linear and angular velocity information over the directed graph $\bar{\mathcal{G}}$. The desired position deviation between the UAV i and the neighbor UAV j can be given by the constant vector $\delta_{ij} \in \mathbb{R}^3$. It gives the desired time-invariant formation patterns. The formation trajectories are time variant and conducted by the virtual leader. The position of the virtual leader $p_0 \in \mathbb{R}^3$ is differentiable, in which the linear velocity of the leader is given by $\dot{p}_0 = v_d$, and the desired position deviation between the UAV i and the leader is given by $\delta_i \in \mathbb{R}^3$, which satisfies $\delta_i - \delta_j = \delta_{ij}$. The control objective is to design distributed control schemes to reach the desired formation patterns and almost-global attitude tracking. Formally, $\forall i = 1, \dots, n$, it is required that

$$\lim_{t \rightarrow \infty} (p_i(t) - p_0(t)) = \delta_i \quad (6)$$

$$\lim_{t \rightarrow \infty} (v_i(t) - v_d(t)) = 0. \quad (7)$$

Instead of the formation tracking with the undirected graph and velocity measurements, we assume the following.

ASSUMPTION 1 In the directed leader-follower interaction topologies $\bar{\mathcal{G}}$, the virtual leader has a directed path that leads to all the follower quadrotors, and the interaction topology of followers $\mathcal{G}$ has a directed spanning tree.

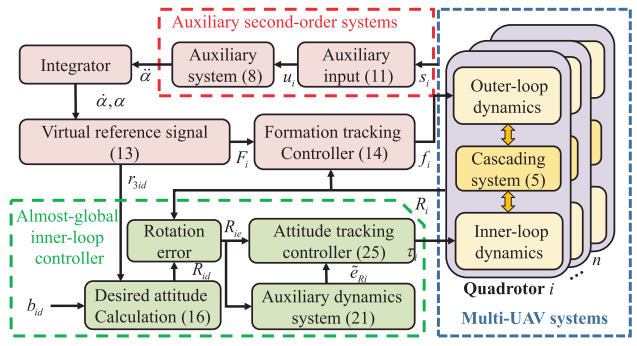


Fig. 2. Distributed formation tracking control architecture for a multiple-quadrotor system without velocity feedback.

ASSUMPTION 2 The trajectory of the leader $p_0(t)$ is sufficiently smooth fourth-order differentiable, and its second to fourth time derivatives all are bounded.

III. MAIN RESULTS

In this section, without employing the linear and angular velocity, we design the distributed formation controller for a multiple-quadrotor system (5) under interaction topologies. To deal with the control challenges caused by the underactuated quadrotors, a hierarchical control framework is developed with the position and attitude loops. Fig. 2 depicts the distributed formation tracking control architecture for a multiple-quadrotor system. At the end of this section, it gives the stability of the overall closed-loop system by using the Lyapunov tool.

A. Formation Tracking Without Linear Velocity Measurements Under a Directed Graph

The formation tracking problem is addressed under the directed interaction topologies. Classically, the distributed control laws for all the followers are designed under an undirected graph to guarantee that each quadrotor has access to the state of its neighbor robots. In particular, the interaction matrix of undirected graphs is symmetric positive definite. Therefore, many interesting results have been obtained based on this excellent property. However, it is not true when interactions of followers are under a directed graph. Under the directed path mentioned in Assumption 1, this section develops a distributed position control law without linear velocity.

First, a second-order auxiliary dynamics system is designed to handle the problem caused by the lack of linear velocity feedback. Define auxiliary variables $\alpha_i \in \mathbb{R}^3$ and $\beta_i \in \mathbb{R}^3$; then, the second-order auxiliary dynamics system is defined as follows:

$$\begin{cases} \ddot{\alpha}_i = -k_{\alpha p} \sigma(\alpha_i) - k_{\alpha d} \sigma(\dot{\alpha}_i) + k_{\beta p} \beta_i + k_{\beta d} \dot{\beta}_i \\ \dot{\beta}_i = -k_{\beta p} \beta_i - k_{\beta d} \dot{\beta}_i - u_i \end{cases} \quad (8)$$

where $k_{\beta p} > 0$ and $k_{\beta d} > 0$ are positive gains. The auxiliary variables $\alpha_i(0)$, $\dot{\alpha}_i(0)$, $\beta_i(0)$, and $\dot{\beta}_i(0)$ are selected arbitrarily. The saturation function σ is defined in Lemma 2.3. The intermediary control input $u_i \in \mathbb{R}^3$ is to be designed later.

REMARK 1 The saturation function plays an important role in the designed auxiliary system. In general, a tanh-type saturation function is commonly used in [21], [22], and [23]. However, any function satisfying Lemma 2.3 can theoretically be employed as a saturation function. To this end, we give the basic definition of the saturation function. It contains almost all the linear or nonlinear saturation functions currently used in similar auxiliary systems.

Denote a new position variable associated with the auxiliary system as

$$\xi_i = p_i - \alpha_i - \beta_i - p_0 - \delta_i. \quad (9)$$

Thus, (6) can be achieved when the translation variable $\xi_i \rightarrow 0$ and the auxiliary dynamic system converges 0.

Under a directed graph, the formation error is given by

$$s_i = \sum_{j=1}^n a_{ij} \xi_{ij} + a_{i0} \xi_i \quad (10)$$

where $\xi_{ij} = \xi_i - \xi_j$. Following the designed formation error, here, the intermediary control input u_i is designed as follows:

$$u_i = -(k_p + k_v)s_i - k_\varepsilon \varepsilon_i \quad (11)$$

where $k_p > 0$ is the positive gain. ε_i used in (11) is arbitrarily initialized and derived solely using position measurements, with no limiting assumptions. ε_i plays the role of a filter at this stage of the control design that is updated by

$$\dot{\varepsilon}_i = -k_\varepsilon \varepsilon_i - k_v s_i \quad (12)$$

where $k_v > 0$ and $k_\varepsilon > 0$ are positive gains.

After u_i is determined, the dynamics of the auxiliary variables β_i and α_i can be obtained by integrating the formula (8) in turn. Furthermore, the reference signal F_i can be obtained based on the auxiliary variable α_i as follows:

$$F_i = -k_{\alpha p} \sigma(\alpha_i) - k_{\alpha d} \sigma(\dot{\alpha}_i) + g e_3 - \dot{v}_d. \quad (13)$$

Reviewing the translational dynamics in (5), the thrust force input f_i for each quadrotor can be obtained as

$$f_i = m_i F_i^T R_i e_3. \quad (14)$$

In addition, the command attitude R_{id} for each quadrotor depends on the reference signal F_i in the cascaded control system (5). Moreover, we can extract the desired unit direction vector $r_{3id} = R_{id} e_3$ from (13) as

$$r_{3id} = -\frac{F_i}{\|F_i\|}. \quad (15)$$

To guarantee that the norm of F_i is nonzero, we assume that the positive gains $k_{\alpha p}$ and $k_{\alpha d}$ are chosen to satisfy

$$\sqrt{3}(k_{\alpha p} + k_{\alpha d}) < g - \max_{t \leq 0} \|\dot{v}_d\|. \quad (16)$$

REMARK 2 To achieve the formation tracking of the under-actuated quadrotors without linear velocity feedback, we design the aforementioned distributed translation control law (14). Specifically, the controller is implemented using the second-order auxiliary dynamics system. It depends on only its current position information and the position

of its neighbor robots and can overcome the lack of unavailable linear velocity. In addition, the designed auxiliary dynamical system is friendly and effective for the directed topologies, so the proposed approach reduces the requirements for communication bandwidth and connectivity in the multiple UAV formations. As a result, the proposed controller can recover and generate the required damping, and one appealing characteristic of our scheme is that it can be easily implemented on low-cost quadrotors.

B. Almost-Global Attitude Tracking Without Angular Velocity Measurements

In this section, the attitude controller for each quadrotor is directly developed by using the rotation matrix. Compared with the Euler angle and unit quaternion, the geometric representation of the rotation motion completely avoids singularities, discontinuities, and ambiguities. Moreover, one of the key ingredients of a geometric controller lies in the definition of the error function. A general framework is based on a real-valued error function. However, it leads to local results with strong initial condition constraints. To relax these constraints, an almost-global controller is designed based on the Lie-algebra-valued functions. Hence, it can offer a convergence region that constitutes the whole state space except a “small” set of measure zero. On this basis, we introduce auxiliary systems to solve the problem of unmeasured angular velocity.

In the following, before we design the attitude controller, an arbitrary smooth attitude tracking command R_{id} for each quadrotor is given. It depends on the reference signal F_i given in (14) and desired yaw angle ψ_{id} . Suppose that ψ_{id} is random; then, the corresponding vector $b_{id} = [\cos \psi_{id}, \sin \psi_{id}, 0]^T$ is not parallel to r_{3id} . Denote $r_{2id} = r_{3id} b_{id} / \|r_{3id} b_{id}\|$, and r_{1id} is selected to be orthogonal to r_{3id} such that $r_{1id} = r_{2id} r_{3id}$. As a result, the desired attitude R_{id} can be denoted by the following exogenous system:

$$R_{id} = [r_{1id}, r_{2id}, r_{3id}] \quad (17)$$

and the associated desired angular velocity and angular acceleration are denoted as

$$\Omega_{id}^\wedge = R_{id}^T \dot{R}_{id}, \quad \dot{\Omega}_{id}^\wedge = (\Omega_{id}^\wedge)^T R_{id}^T \dot{R}_{id} + R_{id}^T \ddot{R}_{id}.$$

To obtain Ω_{id} and $\dot{\Omega}_{id}$, we need to calculate $\dot{R}_{id}$ and $\ddot{R}_{id}$. From (17), it is easily checked that the second-order differentiation of R_{id} can be available when $\dot{F}_i$ and $\ddot{F}_i$ exist. Since $\sigma(\cdot)$ is defined as a continuous function, such as a hyperbolic tangent function, using (13), the second derivative of F_i is given by

$$\begin{aligned} \dot{F}_i &= -k_{\alpha p} \dot{\sigma}(\alpha_i) \dot{\alpha}_i - k_{\alpha d} \dot{\sigma}(\alpha_i) \ddot{\alpha}_i - \ddot{v}_d \\ \ddot{F}_i &= -k_{\alpha p} (\ddot{\sigma}(\alpha_i) \ddot{\alpha}_i + \dot{\sigma}(\alpha_i) \ddot{\alpha}_i) - k_{\alpha d} (\ddot{\sigma}(\alpha_i) \ddot{\alpha}_i \\ &\quad + \dot{\sigma}(\alpha_i) \ddot{\alpha}_i) - \ddot{v}_d. \end{aligned}$$

Based on the second-order auxiliary variables in (8), the third derivative of α_i can be found as

$$\ddot{\alpha}_i = \dot{F}_i + \ddot{v}_d + k_{\beta p} \dot{\beta}_i + k_{\beta d} \ddot{\beta}_i.$$

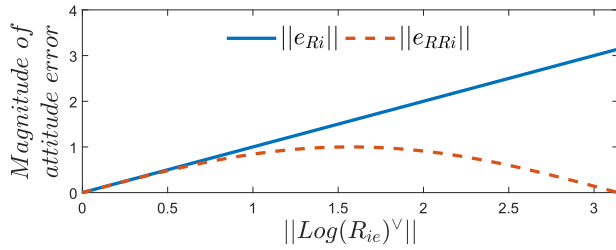


Fig. 3. Magnitude of the attitude error term e_{Ri} defined in (17) and $e_{RRi} = (R_{ie} - R_{ie}^T)^\vee$ defined using a real-valued error function can be obtained over the axis angle $\|(\text{Log}(R_{ie}))^\vee\|$. The magnitude of e_{RRi} is a sine function. In contrast, the magnitude of the proposed attitude error e_{Ri} is a direct proportion function of the axis angle $\|(\text{Log}(R_{ie}))^\vee\|$.

Therefore, we can calculate Ω_d and $\dot{\Omega}_d$ without the linear velocity feedback using the auxiliary variables.

Now, we seek an angular-velocity-free attitude control law τ . Hence, the rotation matrices R_i and the body angular velocity Ω_i can converge to the desired exogenous states R_{id} and Ω_{id} , without using angular velocity information. For the simplification of notation, denote the rotation matrix error as $R_{ie} = R_{id}^T R_i \in \text{SO}(3)$; then, the measure of the rotation belongs to the following open set $U_{(\text{SO}(3))^2}$:

$$U_{(\text{SO}(3))^2} := \left\{ (R_i, R_{id}) \in (\text{SO}(3))^2 : \|I_3 - R_{ie}\| \neq 2\sqrt{2} \right\}.$$

In the spirit of [37], we select the attitude error obtained by the logarithmic map of the rotation matrix error as

$$e_{Ri} = \text{Log}(R_{ie}) \quad (18)$$

where the principal matrix logarithm $\text{Log}(\cdot)$ is defined in Section II-A. It is worth noting that the function $\text{Log}(\cdot) : U_{(\text{SO}(3))^2} \rightarrow \mathfrak{so}(3)$ makes the attitude tracking error e_{Ri} be developed in $\mathfrak{so}(3)$. Furthermore, the differential of $\frac{d}{dt}f(R, R_d)$ concerning $\Omega^\wedge$ is nonsingular on $U_{(\text{SO}(3))^2} \times (\mathfrak{so}(3))^2$. Since $\text{SO}(3)$ is not closed under addition, the distance over $\text{SO}(3)$ cannot be represented in the usual Euclidean sense. Therefore, a smooth feedback control law using tracking error e_{Ri} can achieve almost-global attitude tracking.

REMARK 3 In the existing work [16], the magnitude of the real-valued pose error $e_{RRi} = (R_{ie} - R_{ie}^T)^\vee$ is shown to be a sinusoidal function by a rigorous mathematical derivation. According to the numerical analysis in [14], it is deduced that the attitude error term e_{Ri} is equivalent to the axis-angle rotation on $\mathfrak{so}(3)$. Indeed, Fig. 3 depicts clearly the trend of the two attitude errors' magnitude with the initial angle. The magnitude of the proposed attitude error e_{Ri} is directly proportional to the axis angle $\|(\text{Log}(R_{ie}))^\vee\|$, which means that the convergence speed of e_{Ri} is always faster than that of e_{RRi} . What is more, the convergence speed of e_{RRi} is 0 when the axis angle is equal to $\pi/2$. Therefore, the performance of e_{RRi} becomes worse as the initial attitude error becomes larger. This is not particularly desirable since it destroys one of the distinct advantages of geometric controls, namely, effectiveness for large angle rotational motions. In contrast,

the attitude error not only has the almost-global convergence region but also performs well in convergence speed.

Since $\frac{d}{dt}(\text{Log}(R_i)) = (R_i^T)^\vee \dot{R}_i = \Omega_i^\wedge$, taking the derivative of (19), its associated angular velocity error is given by

$$e_{\Omega_i} = \Omega_i^\wedge - R_{ie}^T \Omega_{id}^\wedge R_{ie}. \quad (19)$$

Note that in (19), angular velocity Ω_i is required to implement the control algorithm. To this end, the rotational auxiliary dynamics system is designed as follows:

$$\dot{\bar{R}}_{ie} = \bar{R}_{ie} \bar{\Omega}_{ie}^\wedge \quad (20)$$

where $\bar{R}_{ie}$ is an auxiliary rotation matrix, and $\bar{\Omega}_{ie}$ is an auxiliary system input to be designed later. Let $\tilde{R}_{ie} = \bar{R}_{ie}^T R_{ie} \in \text{SO}(3)$ be a new error to represent the discrepancy between R_{ie} and $\bar{R}_{ie}$. Similar to (18), the auxiliary error $\tilde{e}_{Ri} \in \mathfrak{so}(3)$ is given by

$$\tilde{e}_{Ri} = \text{Log}(\tilde{R}_{ie}). \quad (21)$$

Then, the following auxiliary system input is designed:

$$\bar{\Omega}_{ie} = k_e \tilde{R}_{ie} \tilde{e}_{Ri}^\vee \quad (22)$$

where $k_e > 0$ is a positive gain.

Recalling the quadrotor dynamics (5), we can obtain the derivative of the angular velocity error as

$$\begin{aligned} J \dot{e}_{\Omega_i}^\vee &= \tau_i - (e_{\Omega_i} + R_{ie}^T \Omega_{id}^\wedge R_{ie}) J (e_{\Omega_i} + R_{ie}^T \Omega_{id}^\wedge R_{ie})^\vee \\ &\quad + J e_{\Omega_i} R_{ie}^T \Omega_{id} - J R_{ie}^T \dot{\Omega}_{id}. \end{aligned} \quad (23)$$

Exploiting the defined error e_{Ri} and $\tilde{e}_{Ri}$, we design an almost-global attitude controller for each quadrotor as follows:

$$\tau_i = -k_R e_{Ri}^\vee - k_{\tilde{R}} \tilde{e}_{Ri}^\vee + J R_{ie}^T \dot{\Omega}_{id} + (R_{ie}^T \Omega_{id})^\wedge J (R_{ie}^T \Omega_{id}) \quad (24)$$

where $k_R > 0$ and $k_{\tilde{R}} > 0$ are positive gains.

REMARK 4 The almost-global geometric attitude controller has been developed directly using the rotation matrix on the special Euclidean group. Thus, the geometric attitude representation can completely avoid the singularities and ambiguities associated with Euler angles or quaternions. Different from [32], the attitude error is defined in the Lie algebra $\mathfrak{so}(3)$ to achieve an almost-global convergence region. More importantly, we design an auxiliary system based on Lie algebra to overcome the lack of angular velocity feedback. In addition, it is easy to note that e_{Ri} is not defined globally when $R_{ie} = \text{diag}(-1, -1, 1)$, $R_{ie} = \text{diag}(-1, 1, -1)$, and $R_{ie} = \text{diag}(1, -1, -1)$. The controller offers a convergence region that constitutes the whole state space except for a “small” set of measure zero. Owing to the presence of topological obstructions, no geometric control law is able to globally track the command attitude. Therefore, the proposed geometric control law is the strongest possible controller with respect to the convergence region.

C. Stability Analysis

THEOREM 1 Under a directed follower graph $\mathcal{G}$ mentioned as Assumption 1, using the formation tracking controller (14) and attitude tracking controller (24) for a multi-UAV system (5), the quadrotor team achieves the desired formation specified in (6) and (7) if and only if Condition 1 holds.

CONDITION 1 The gains of formation tracking controller are chosen to satisfy

$$\frac{k_v k_\varepsilon^2 c_i}{(k_p + k_v)^2} > \max d_i^2 \quad (25)$$

$$\min \{k_p k_v^2 k_\varepsilon^3 c_i^3 - k_p k_\varepsilon (k_p + k_v)^3 d_i^2 (c_i^2 + d_i^2) + k_p k_\varepsilon^3 (k_v^2 - 2k_p^2 - k_p k_v) d_i^2 c_i + k_\varepsilon^5 k_p^2 d_i^2\} > 0 \quad (26)$$

where $\vartheta_i = c_i + j d_i$, $i = 1, 2, \dots, n$, are the eigenvalues of matrix $\mathcal{H}$, and c_i and d_i are the real numbers.

PROOF In what follows, we proceed in three steps to prove the stability of the closed-loop system.

Step 1: The stability of the attitude error dynamics.

Using the attitude kinematics equations $\dot{R}_i = R_i \Omega_i^\wedge$ and prior definition in (20), we have $\dot{R}_{ie} = R_{ie} e_{\Omega_i}$ and $\dot{\tilde{R}}_{ie} = \tilde{R}_{ie} \tilde{\Omega}_{ie}^\wedge$; then

$$\begin{aligned} \dot{\tilde{R}}_{ie} &= \tilde{R}_{ie}^T \tilde{R}_{ie} e_{\Omega_i} - \tilde{\Omega}_{ie}^\wedge \tilde{R}_{ie} = \tilde{R}_{ie} e_{\Omega_i} - \tilde{R}_{ie} \tilde{R}_{ie}^T \tilde{\Omega}_{ie}^\wedge \tilde{R}_{ie} \\ &= \tilde{R}_{ie} (e_{\Omega_i}^\vee - \tilde{R}_{ie}^T \tilde{\Omega}_{ie}^\wedge)^\wedge. \end{aligned} \quad (27)$$

Substituting auxiliary system input (22) into (27), one obtains

$$\dot{\tilde{R}}_{ie} = \tilde{R}_{ie} (e_{\Omega_i} - k_e \tilde{e}_{Ri}). \quad (28)$$

Using (18), (21), (23), and (28), the attitude error dynamics is calculated as follows:

$$\dot{e}_{Ri} = e_{\Omega_i} \quad (29)$$

$$\begin{aligned} J \dot{e}_{\Omega_i}^\vee &= -k_R e_{Ri}^\vee - k_R \tilde{e}_{Ri}^\vee - e_{\Omega_i} J (e_{\Omega_i}^\vee + R_{ie}^T \Omega_{id}) \\ &\quad - (R_{ie}^T \Omega_{id})^\wedge J e_{\Omega_i}^\vee + J e_{\Omega_i} R_{ie}^T \Omega_{id} \end{aligned} \quad (30)$$

$$\dot{\tilde{e}}_{Ri} = \tilde{R}_{ie}^T \tilde{R}_{ie} = e_{\Omega_i} - k_e \tilde{e}_{Ri}. \quad (31)$$

Define the Lyapunov candidate function as

$$V_1 = \frac{k_R}{2} (e_{Ri}^\vee)^T e_{Ri}^\vee + \frac{1}{2} (e_{\Omega_i}^\vee)^T J e_{\Omega_i}^\vee + \frac{k_R}{2} (\tilde{e}_{Ri}^\vee)^T \tilde{e}_{Ri}^\vee. \quad (32)$$

Next, we take the derivative of (32) as follows:

$$\begin{aligned} \dot{V}_1 &= (e_{Ri}^\vee)^T k_R \dot{e}_{Ri}^\vee + (e_{\Omega_i}^\vee)^T J \dot{e}_{\Omega_i}^\vee + (\tilde{e}_{Ri}^\vee)^T k_R \dot{\tilde{e}}_{Ri}^\vee \\ &= k_R (e_{Ri}^\vee)^T e_{\Omega_i}^\vee + (e_{\Omega_i}^\vee)^T \{ -k_R e_{Ri}^\vee - k_R \tilde{e}_{Ri}^\vee \\ &\quad - e_{\Omega_i} J (e_{\Omega_i}^\vee + R_{ie}^T \Omega_{id}) - (R_{ie}^T \Omega_{id})^\wedge J e_{\Omega_i}^\vee \\ &\quad + J e_{\Omega_i} R_{ie}^T \Omega_{id} \} + (\tilde{e}_{Ri}^\vee)^T (e_{\Omega_i}^\vee - k_e \tilde{e}_{Ri}^\vee) \\ &= -k_e (\tilde{e}_{Ri}^\vee)^T \tilde{e}_{Ri}^\vee - (e_{\Omega_i}^\vee)^T e_{\Omega_i} J (e_{\Omega_i}^\vee + R_{ie}^T \Omega_{id}) \\ &\quad + (e_{\Omega_i}^\vee)^T \left[- (R_{ie}^T \Omega_{id})^\wedge J + J (R_{ie}^T \Omega_{id})^\wedge \right] (e_{\Omega_i}^\vee). \end{aligned} \quad (33)$$

Since the skew matrix e_{Ω_i} satisfies $e_{\Omega_i} e_{\Omega_i}^\vee = 0$ and $\left[- (R_{ie}^T \Omega_{id})^\wedge J + J (R_{ie}^T \Omega_{id})^\wedge \right]$ is skew symmetric, the last two terms in (33) become zero; then, we have

$$\dot{V}_1 = -k_e \|\tilde{e}_{Ri}^\vee\|^2 \leq 0. \quad (34)$$

Therefore, the attitude dynamics can be guaranteed to be stable on $U(\text{SO}(3))^2 \times \mathfrak{so}(3)^2$ by using Lyapunov tools. Since V_1 is bounded, then $e_{Ri}^\vee$, $e_{\Omega_i}^\vee$, and $\tilde{e}_{Ri}^\vee$ are bounded. From (34), one gives the boundedness of $\tilde{e}_{Ri}^\vee$. Exploiting Barbalat's lemma, $\tilde{e}_{Ri}^\vee \rightarrow 0$ as $t \rightarrow \infty$. Since $e_{\Omega_i}^\vee$, $\dot{e}_{\Omega_i}^\vee$, $\tilde{e}_{Ri}^\vee$, and $\dot{\tilde{e}}_{Ri}^\vee$ are bounded, then one gets that $\tilde{e}_{Ri}^\vee$ is bounded. Based on the Barbalat's lemma, $\tilde{e}_{Ri}^\vee \rightarrow 0$; then, $e_{\Omega_i}^\vee \rightarrow 0$. Because Ω_{id} , $\Omega_{id}^\vee$, $e_{Ri}^\vee$, and $\dot{e}_{Ri}^\vee$ are bounded, $\dot{e}_{\Omega_i}^\vee$ is bounded. Exploiting Barbalat's lemma, $e_{\Omega_i}^\vee \rightarrow 0$, and bounded $\tilde{e}_{Ri}^\vee$ gives $\dot{e}_{\Omega_i}^\vee \rightarrow 0$, and hence, $e_{Ri}^\vee \rightarrow 0$ as $t \rightarrow \infty$.

Step 2: The stability of the position error dynamics.

We start by reviewing the position variables ξ_i and the input of the filter-like auxiliary system (8)–(12). Furthermore, they can be used to characterize the closed-loop form of the multi-UAV system. Denote $\eta_i = \dot{\xi}_i = v_i - \dot{\alpha}_i - \dot{\beta}_i - v_d$. Based on the control input in (11), the derivative of η_i yields the following result:

$$\begin{aligned} \dot{\eta}_i &= F_i - \ddot{\alpha}_i - \ddot{\beta}_i - \dot{v}_d + (\ddot{p}_i - F_i) \\ &= -k_p (s_i - \dot{\xi}_i) + \varpi_i \end{aligned} \quad (35)$$

and

$$\varpi_i = \frac{f_i}{m_i e_3^T R_{id}^T R_{ie} e_3} ((e_3^T R_{id}^T R_{ie} e_3) R_{ie} e_3 - R_{id} e_3) \quad (36)$$

where ϖ_i is a nonlinear interconnection term. It specifically presents the high coupling of the quadrotor system (5).

Define vectors $\xi = [\xi_1^T, \xi_2^T, \dots, \xi_n^T]^T$, $\eta = [\eta_1^T, \eta_2^T, \dots, \eta_n^T]^T$, $\varepsilon = [\varepsilon_1^T, \varepsilon_2^T, \dots, \varepsilon_n^T]^T$, and $\varpi = [\varpi_1^T, \varpi_2^T, \dots, \varpi_n^T]^T$. In addition, denote $X = [\xi^T, \dot{\xi}^T, \varepsilon^T]^T$ and $g(X(t), t) = [0_n^T, \varpi^T, 0_n^T]^T$. Then, we have the following matrix form:

$$\dot{X} = (\Gamma \otimes I_3)X + g(X, t) \quad (37)$$

where the state transformation matrix Γ is given by

$$\Gamma = \begin{bmatrix} 0_n & I_n & 0_n \\ -(k_p + k_v)\mathcal{H} & 0_n & -k_\varepsilon \otimes I_n \\ -k_v \mathcal{H} & 0_n & -k_\varepsilon \otimes I_n \end{bmatrix}. \quad (38)$$

Since signals F_i and f_i are bounded based on Assumption 2 and Lemma 2.3, it is noted that $\varpi_i \rightarrow 0$ when $e_{Ri} \rightarrow 0$. In this step, assuming the coupling stem $g(X, t) = 0$, one gives

$$\dot{X} = (\Gamma \otimes I_3)X. \quad (39)$$

Next, we can analyze the stability of system (38) by determining checking the eigenvalues of Γ . To find the eigenvalues of Γ , we can solve the equation $\det(\lambda I_{3n} - \Gamma) = 0$, and the characteristic polynomials $\det(\lambda I_{3n} - \Gamma)$ can be

given by

$$\begin{aligned} & \det(\lambda I_{3n} - \Gamma) \\ &= \det \left(\begin{bmatrix} \lambda I_n & -I_n & 0_n \\ (k_p + k_v)\mathcal{H} & \lambda I_n & k_\varepsilon \otimes I_n \\ k_v \mathcal{H} & 0_n & \lambda I_n + k_\varepsilon \otimes I_n \end{bmatrix} \right) \\ &= \det [\lambda^3 I_n + k_\varepsilon \lambda^2 I_n + (k_p + k_v)\lambda \mathcal{H} + k_p k_\varepsilon \mathcal{H}]. \end{aligned}$$

Note that

$$\det(\lambda I_n - \mathcal{H}) = \prod_i^n [\lambda - \vartheta_i] \quad (40)$$

where $\vartheta_i = c_i + j d_i$, $i = 1, 2, \dots, n$, are the eigenvalues of $\mathcal{H}$.

As a result, the characteristic polynomials of Γ can be written as

$$\det(\lambda I_{3n} - \Gamma) = \prod_i^n G_i(\lambda) \quad (41)$$

where

$$\begin{aligned} G_i(\lambda) &= \lambda^3 + k_\varepsilon \lambda^2 + [(k_p + k_v)c_i + j(k_p + k_v)d_i]\lambda \\ &\quad + k_p k_\varepsilon c_i + j k_p k_\varepsilon d_i. \end{aligned}$$

Under the light of Lemma 2.2, if and only if Condition 1 is satisfied, all three roots of $G_i(\lambda) = 0$ have negative real parts. Therefore, the system (37) will be exponentially asymptotically stable. Then, we have that $\xi_i \rightarrow 0$, $\ddot{\xi}_i \rightarrow 0$, and $\dot{\varepsilon}_i \rightarrow 0$ as $t \rightarrow \infty$; then, we can get from the linear system (42) that $\lim_{t \rightarrow \infty} \xi_i(t) = 0$ and $\lim_{t \rightarrow \infty} \varepsilon_i(t) = 0$; thus, we can verify that u_i asymptotically converges to be zero. Then, applying Lemma 2.5 to the auxiliary system (11) in sequence, we have $\alpha_i(t)$, $\dot{\alpha}_i(t)$, $\beta_i(t)$, and $\dot{\beta}_i(t)$ tend to zero. As a result, $\varphi_i \rightarrow 0$, $\alpha_i \rightarrow 0$, and $\beta_i \rightarrow 0$ yield $\lim_{t \rightarrow \infty} (p_i(t) - p_0(t)) = \delta_i$.

Step 3: The stability of the overall system with the coupling term. In this step, we investigate the stability of the overall system with coupled term $g(X, t)$. Denote $A = \Gamma \otimes I_3$, the overall system is presented as follows:

$$\dot{X} = AX + g(X, t). \quad (42)$$

We define the Lyapunov functional V_2 as

$$V_2 = X^T P X \quad (43)$$

where P is a positive-definite matrix. From Step 2, it has been proven that the eigenvalues of system matrix A have negative real parts under Condition 1. Hence, there is a positive-definite matrix Q satisfies $A^T P + PA = -Q$ so that we get

$$\dot{V}_2 = -X^T Q X + 2X^T P g(X, t). \quad (44)$$

Taking the modulo of the last term of (44), one obtains

$$\|2X^T P g(X, t)\| \leq \|2P g(X, t)\| \|X\|. \quad (45)$$

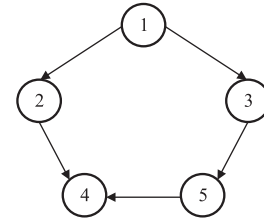


Fig. 4. Directed topology of five quadrotors.

Denoting $\gamma = \|2P g(X, t)\| / \lambda_{\min}(P)$, one obtains

$$\dot{V}_2 \leq -W_2 + \gamma \sqrt{V_2} \quad (46)$$

where $W_2 = \frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} V_2$. Since signals F_i and f_i are bounded based on Assumption 2, the coupling term ϖ_i can be proven to be bounded and the detailed proof steps can be found in [16], so it can be easily checked that $g(X, t)$ is bounded for all time. The attitude dynamics is asymptotically stable and $e_{Ri}^\vee \rightarrow 0$, so we have $\varpi_i \rightarrow 0$, $g(X, t) \rightarrow 0$, and $\gamma \rightarrow 0$. Therefore, it can be seen that V_2 satisfies the conditions in Lemma 2.4 such that $V_2(t) \rightarrow 0$ as $t \rightarrow \infty$, and the overall system is asymptotically stable. This completes the proof. ■

REMARK 5 Condition 1 is necessary to guarantee the stability of the multiunderactuated system in an undirected interaction topology. In particular, it is easy to implement and satisfies $d_i = 0$ for undirected graphs. Therefore, the proposed controller not only solves velocity-free formations in directed graphs but also works in undirected graphs.

IV. SIMULATION RESULTS

In this section, several simulations are conducted to show the effectiveness and performance of the proposed formation tracking algorithm. No prior linear and angular velocities of each quadrotor are provided. Consider a multi-vehicle system (5) composed of $n = 5$ followers with a virtual leader under the directed network topology shown in Fig. 4. Moreover, let $a_{ij} = 1$ or 0 depending on the directed graph. The parameters of each quadrotor were given as $J_i = \text{diag}(0.5, 0.5, 0.5) \text{ kg} \cdot \text{m}^2/\text{rad}^2$ and $m_i = 1 \text{ kg}$ for $i = 0, 1, 2, 3, 4$. The desired formation pattern specified by the (6) was chosen as $\delta_{10} = [-1.6, -1.2, 0]^T$, $\delta_{20} = [-1, -3, 0]^T$, $\delta_{30} = [1, -3, 0]^T$, $\delta_{40} = [1.6, -1.2, 0]^T$, and $\delta_{50} = [1.6, -1.2, 0]^T$. The desired yaw angle for each quadrotor was given by $\psi_d = 90^\circ$. The initial rotation matrix was given by $R_i(0) = I_3$. The auxiliary variables $\alpha_i(0)$, $\dot{\alpha}_i(0)$, $\beta_i(0)$, and $\dot{\beta}_i(0)$ were initialized zero. In addition, let $\varepsilon_i(0) = 0$. In addition, the parameters of the formation controllers were chosen to satisfy Condition 1 as Table I. Using the controllers (14) and (24) to generate the saturated thrust and torque for each quadrotor. In the following, a regular pentagonal formation pattern is given with the following three cases.

Case 1: The velocity of the virtual leader is constant.

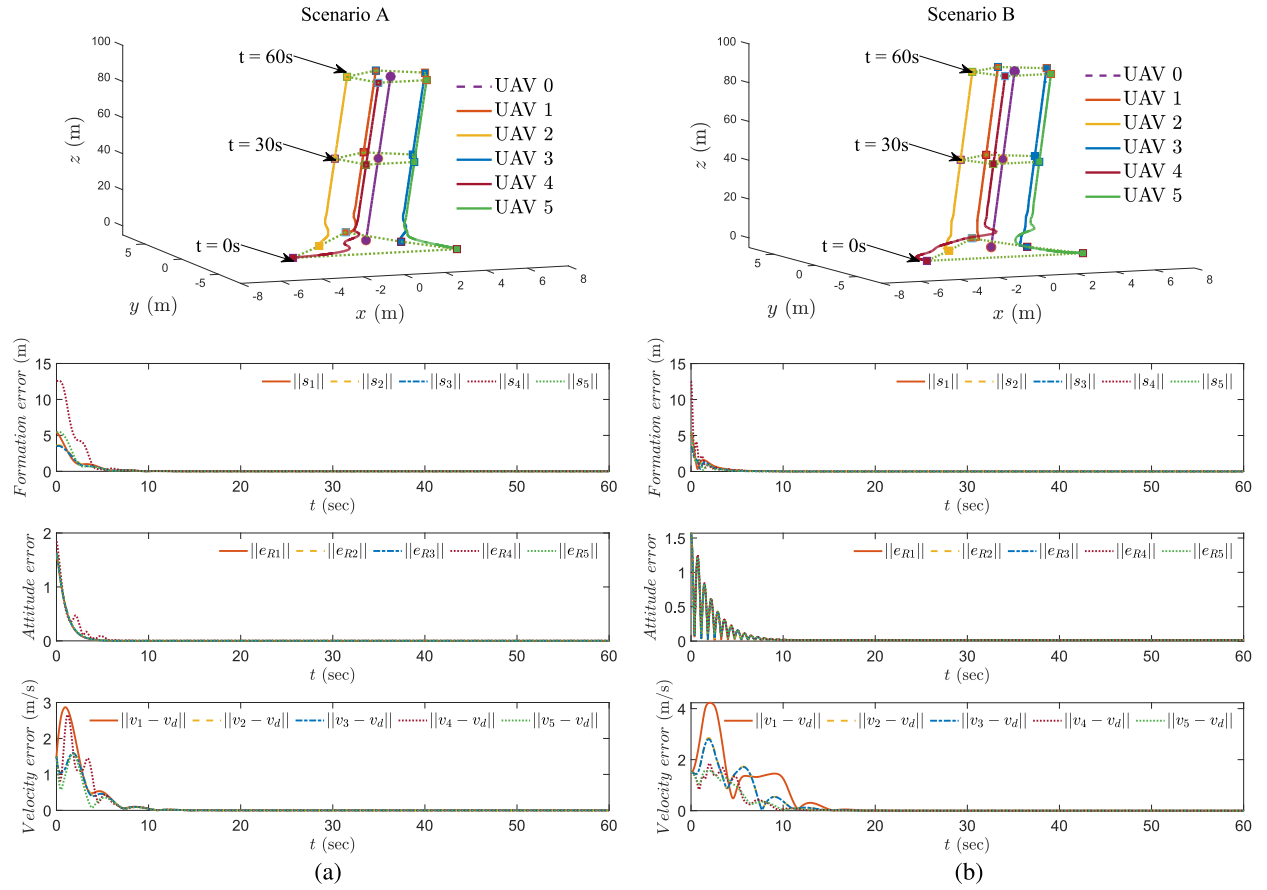


Fig. 5. Simulation case 1: Quadrotor formation tracking under scenarios A and B. (a) Formation trajectories, formation tracking errors, attitude errors, and linear velocity errors under scenario A. (b) Formation trajectories, formation tracking errors, attitude errors, and linear velocity errors under scenario B.

TABLE I
Parameters of Quadrotor Tracking Control

Parameters	Value	Parameters	Value
k_p	1	k_v	10
k_e	5	$k_{\alpha p}$	1
$k_{\alpha d}$	1	$k_{\beta p}$	1
$k_{\beta d}$	1	k_e	2
k_R	5	$k_{\bar{R}}$	5

Case 2: The velocity of the virtual leader is time-varying.

Case 3: A comparative study to achieve the aggressive formation maneuvers with the local attitude controller.

To test the control performance of the proposed velocity-free formation controller with respect to the algorithm with velocity measurement, two different formation scenarios are specified.

- Ideal formation environment, in which the linear and angular velocities of the quadcopter are assumed to be accurately available, uses the algorithm with velocity measurements.
- Harsh formation environments, where the velocities are not measurable, use the proposed velocity-free formation controller.

The reference linear velocity of the virtual group leader was given by $v_d = [0.02, 0, 1.5]^T$. The results are shown in Fig. 5. Fig. 5(a) depicts the formation trajectories, formation tracking errors, attitude errors, and linear velocity errors under scenario A. Fig. 5(b) shows the formation trajectories, formation tracking errors, attitude errors, and linear velocity errors under scenario B. Compared to the transient performance in scenario A, the overshoot and oscillation of the proposed velocity-free controller are slightly larger during convergence due to the use of a filter-like approach to address the unknown velocities. However, it can be seen that the quadrotors achieve the desired formation patterns well in the absence of linear and angular velocity feedback in scenario B. This implies that the formation reliability against limited sensing resources and sensor failures in practice is well enhanced. Therefore, these studies show the superior robustness of the proposed approach over the comparative method in handling inaccurate velocity measurements and sensor failures.

In the second case, the performance of the formation tracking controller with time-varying reference velocity was evaluated. The reference linear velocity of the virtual group leader with complex motion was given by $v_d = [0.5 \sin(0.2t), 0.5 \cos(0.2t), 3]^T$. The results are shown in Fig. 6. In Fig. 6(a), the 3-D formation trajectory of a group of quadrotors is displayed for 60 s. Fig.

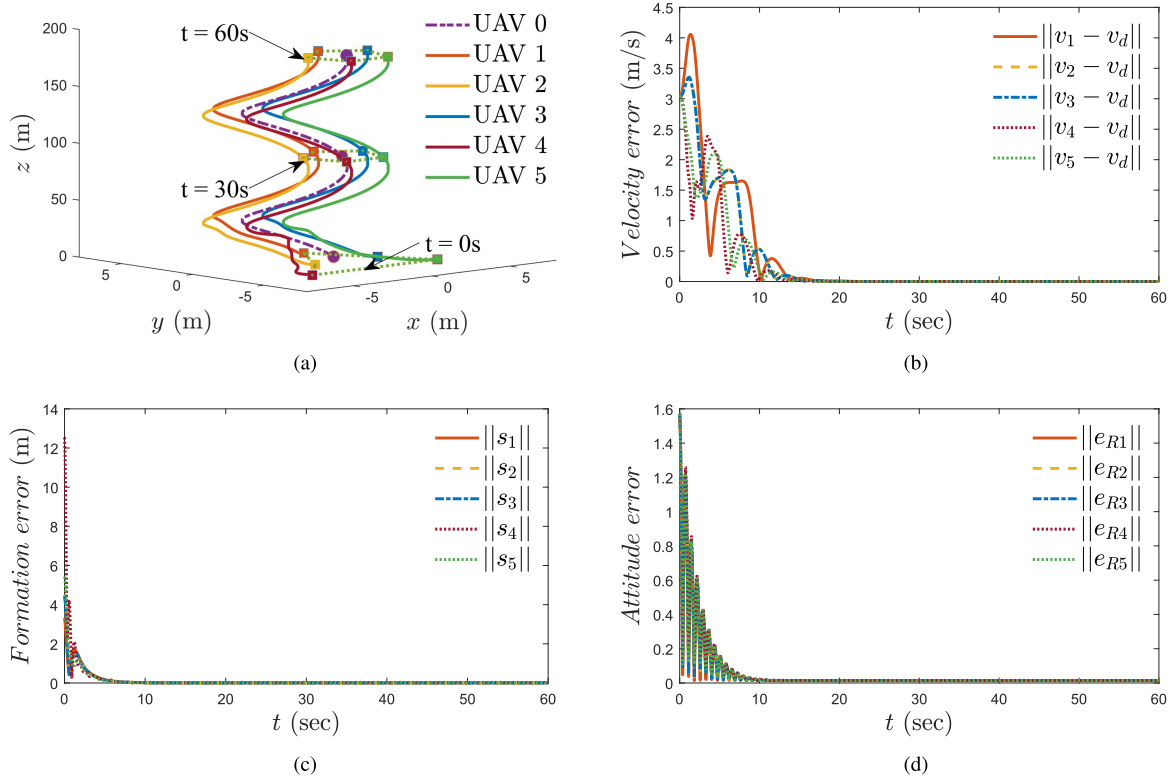


Fig. 6. Simulation case 2: Quadrotor formation tracking, where the velocity of the virtual leader is time varying. (a) Trajectories of the virtual group leader UAV 0 (purple line), and trajectories of a group of follower quadrotors in the 3-D space. (b) Time response of linear velocity errors. (c) Time response of formation tracking errors. (d) Attitude errors for each follower quadrotor.

6(b) depicts the linear velocity errors. Fig. 6(c) and (d) shows the formation and attitude errors of each follower quadrotor. From the perspective of tracking performance, the proposed distributed formation controller can ensure that each follower smoothly tracks a leader with the desired pattern in a short time.

In the third case, under the same simulation conditions, we evaluated the proposed attitude controller without angular velocity feedback and compared the results with those of the controller developed in [32]. The attitude tracking scheme in [32] was similarly defined for the desired attitude constructed in (17) and dealt with the unavailable angular velocity. The attitude controller in [32] was local since the attitude error vector was defined as $e_{Ri} = (R_{ie} - R_{ie}^T)^\vee$ using the real-valued error functions. In contrast, the proposed almost-global attitude error vector was defined as $e_{Ri} = \text{Log}(R_{ie})$ using the Lie-algebra-valued functions. Because the attitude controller in [32] has the local region of convergence where the rotation angle between R_i and R_{id} is less than 180° , to verify the almost-global performance of the proposed attitude control law, we investigated the situations where the initial rotation angle between $R_i(0)$ and R_{id} is equal to 170° and 180° . The linear velocity of the virtual leader was given by $v_d = [0.2 \sin(0.1t), 0.2 \cos(0.1t), 1]^T$. The simulation results are

depicted in Fig. 7, in which snapshots of quadrotor positions are provided every 30 s. In addition, the directions of each UAV at 0 and 60 s are given, indicated by a red arrow.

The comparative results are illustrated in Fig. 7(a) and (b) under a relatively aggressive initial rotation angle of 170° . It can be seen that the formation trajectory of the proposed approach is smoother, while the comparative method has a large shock and is affected by the attitude error. For another, the proposed method has a faster convergence speed, which verifies the transient performance of the proposed controller mentioned in Remark 3. The results show that the comparative attitude controller is subject to a relatively strong initial condition constraint. Hence, the performance of the attitude controller decreases significantly under a relatively aggressive initial rotation angle. In particular, the attitude error e_{Ri} is completely invalidated when the initial rotation angle between $R_i(0)$ and R_{id} is equal to 180° . In contrast, the proposed method still performs well under this extreme initial condition; the simulation results are shown in Fig. 7(c). The tracking trajectory and the errors change are similar to those in Fig. 7(b). As a result, we can confirm that the proposed almost-global controller is capable of performing aggressive formation tracking even under such extreme initial conditions. Based on the above

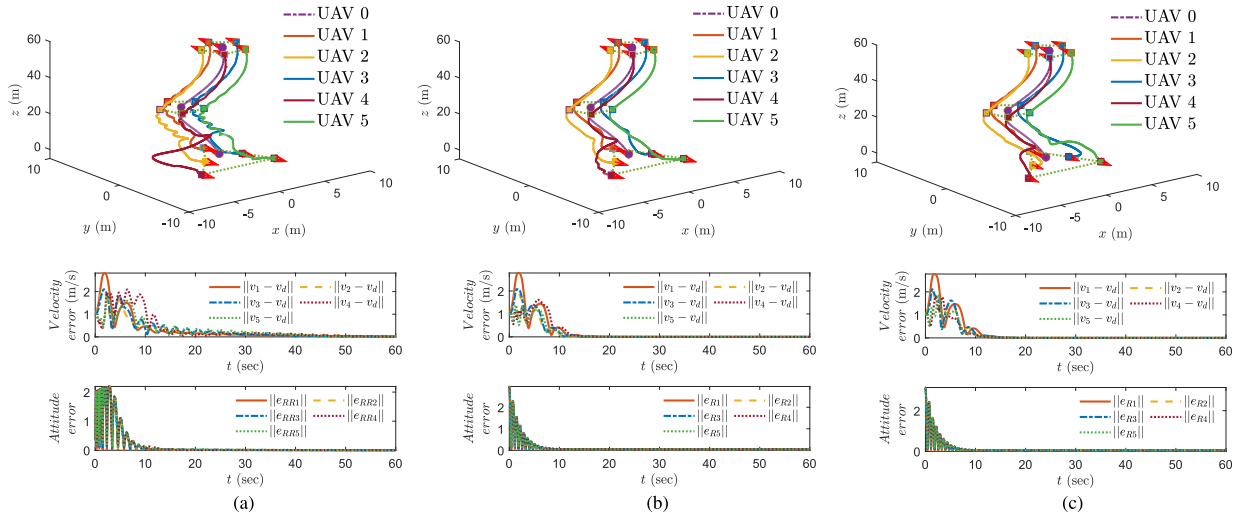


Fig. 7. Simulation case 3: A comparative study to achieve the aggressive formation maneuvers with the local attitude controller. (a) Tracking performance of the comparative method under the initial rotation angle of 170° . (b) Tracking performance of the proposed method under the initial rotation angle of 170° . (c) Tracking performance of the proposed method under the initial rotation angle of 180° .

analysis, it can be concluded that the proposed attitude controller outperforms the local attitude controller in terms of both the convergence region and the convergence speed.

V. CONCLUSION

In this article, we presented a distributed formation tracking scheme for a group of low-cost quadrotor without velocity measurements over directed topologies. To handle the control challenges caused by the underactuated nature of the quadrotor, a hierarchical control framework was developed with the position and attitude loops. In the position loop, a second-order auxiliary system was designed to overcome the unmeasurable linear velocity. Therefore, the translation controller was designed using only the position information under the directed topologies. In the attitude loop, an almost-global geometric attitude controller was proposed to carry out aggressive rotation maneuvers. What is more, an auxiliary system based on Lie-algebra-valued functions was designed to eliminate the unreliable angular velocity feedback. The stability analysis of the whole closed-loop error system was analyzed via the Lyapunov stability theory. Numerical simulations were shown to demonstrate the effectiveness and performance of the designed formation control scheme. Future work will focus on improving the safe operation of UAVs considering collision avoidance.

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